CSCI E-106:Assignment 3

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Problem 1

Refer to the Crime rate data. A criminologist studying the relationship between level of education-and crime rate in medium-sized U.S. counties collected the following data for a random sample of 84 counties; X is the percentage of individuals in the county having at least a high-school diploma, and Y is the crime rate (crimes reported per 100,000 residents) last year. (100 points)

```r
set.seed(10)
crime_rate_data <- read.csv("Crime Rate.csv")
```

a-) Create train data set by selecting 70% of the data randomly and use the remaining 30% of the data as a test data set (use set.seed(10) before selecting the sample).

```r
popN = nrow(crime_rate_data)
alphaN = 7/10 * popN
idx_sample = sample(popN, size = alphaN, replace=FALSE)
sample = crime_rate_data[idx_sample,]
sampleN = nrow(sample)
test = crime_rate_data[-idx_sample,]
testN = nrow(test)

training_data = sample
test_data = test
```

Use train data set to answer questions for part b to i

b-) *Build a regression model to predict Y on the train data set.*

```r
linear_model = lm(Y ~ X, training_data)
summary(linear_model)
```

```
##
## Call:
## lm(formula = Y ~ X, data = training_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4905.5 -1825.3  -443.6  1803.8  7136.3
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 17080.68    4139.62   4.126 0.000124 ***
## X            -130.78      52.67  -2.483 0.016051 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2419 on 56 degrees of freedom
## Multiple R-squared:  0.09917,    Adjusted R-squared:  0.08309
```

```
## F-statistic: 6.165 on 1 and 56 DF,  p-value: 0.01605
```

Comment on your model.

The p-value for the intercept is $0.000124 < 0.05$. This indicates that the intercept is statistically significant.

The p-value for the slope is $0.016051 < 0.05$. This indicates that the slope is statistically significant.

The F-statistic is $6.165 >> 1$. We would reject the hypothesis that the slope is 0.

Are there any outliers in the data set?

We can assess whether the errors are normally distributed by looking for outliers in the residuals, typically, errors more than $2 * \sigma$ are considered moderate outliers.

```r
print(linear_model$residuals[linear_model$residuals / sigma(linear_model) > 2])
```

```
##       27
## 7136.252
```

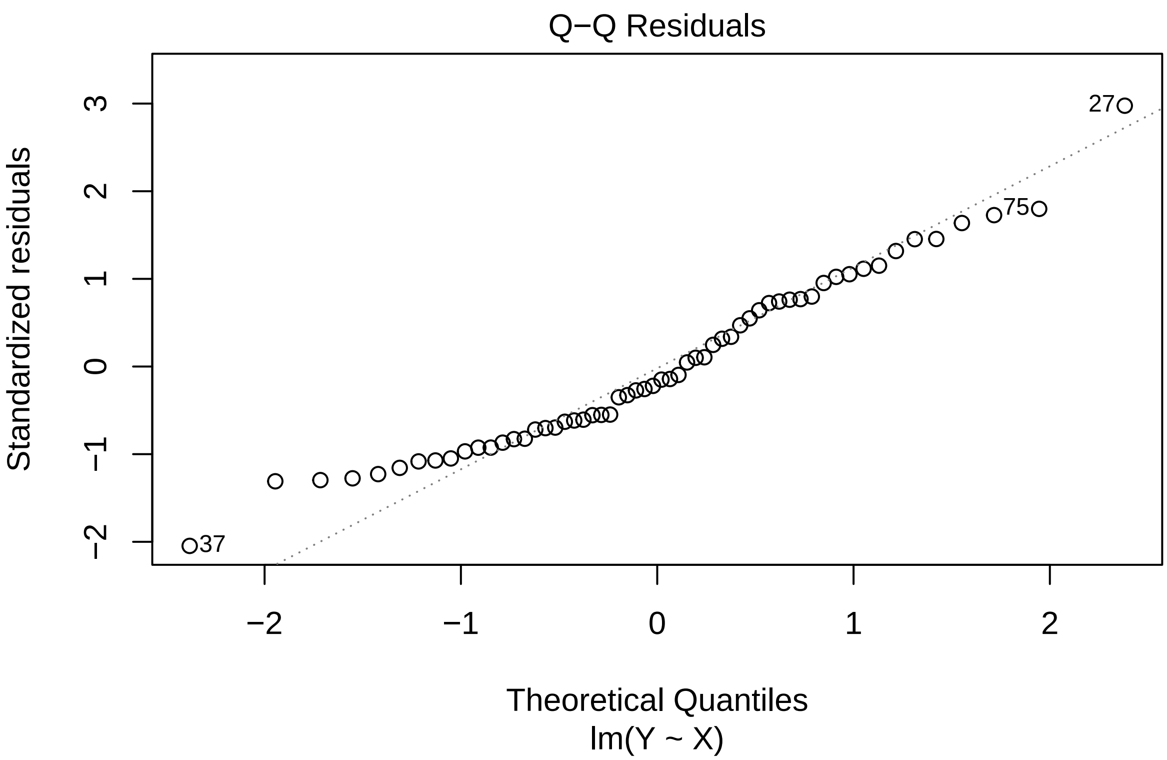
A single outlier is identified. The rest of the residuals tend to fit a standard normal distribution.

Are there errors normally distributed?

Q-Q plot check for normally distributed residuals.

We can look at the Q-Q plot of the residuals. If the residuals are normally distributed, they will fall into a straight line.

```r
plot(linear_model, which = c(2))
```



The residuals appear to deviate at the ends of the plot, suggesting they are not normally distributed.

Shapiro Test for normally distributed residuals.

We can run a Shapiro test to get a P-value. The null hypothesis is that the data follow the normal distribution. The alternative hypothesis is that the data do not follow the normal distribution.

```
lm_res = linear_model$residuals
st = shapiro.test(lm_res)
st
```

```
##
##  Shapiro-Wilk normality test
##
## data:  lm_res
## W = 0.96489, p-value = 0.09147
```

The p-value for Shapiro is 0.0914664.

```
shapiro_decision = st$p.value > 0.05
if (st$p.value > 0.05) {
  print(sprintf("%.3f > 0.05 : H_0", st$p.value))
} else {
  print(sprintf("%.3f < 0.05 : H_a", st$p.value))
}
```

```
## [1] "0.091 > 0.05 : H_0"
```

Kolmogorov-Smirnov Test for normally distributed residuals.

We can run a Kolmogorov-Smirnov test to get a P-value. The null hypothesis is that the data follow the normal distribution. The alternative hypothesis is that the data do not follow the normal distribution.

```
kst = ks.test(lm_res, "pnorm", mean(lm_res), sd(lm_res))
kst
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  lm_res
## D = 0.12203, p-value = 0.3266
## alternative hypothesis: two-sided
```

The p-value for K.S. is 0.3266293.

```
if (kst$p.value > 0.05) {
  print(sprintf("%.3f > 0.05, so we fail to reject the null hypothesis.", kst$p.value))
} else {
  print(sprintf("%.3f < 0.05, so we accept the alternative hypothesis.", kst$p.value))
}
```

```
## [1] "0.327 > 0.05, so we fail to reject the null hypothesis."
```

c-) *Set up the ANOVA table for the regression model.*

Let's state the null and alternative hypotheses:

$$H_0 : \beta_1 = 0$$
$$H_a : \beta_1 \neq 0$$

```
anova_table_constructor <- function(training_data) {
    linear_model <- lm(training_data)
```

```r
    mean_y = mean(training_data[,"Y"])

    r_h0 = mean_y - training_data[,"Y"]
    r_h1 = linear_model$residuals
    r_r = predict(linear_model, training_data) - mean_y

    anova_table = data.frame(row.names = c("Regression", "Error", "Total"))
    anova_table = cbind(anova_table, "D.F."=c(
      1,
      nrow(training_data)-2,
      nrow(training_data)-1)
    )
    anova_table = cbind(anova_table, "S.S."=c(
      sum(r_r^2),
      sum(r_h1^2),
      sum(r_h0^2))
    )
    anova_table = cbind(anova_table, "M.S.S."=
                          c(anova_table[,"S.S."]/anova_table[,"D.F."])
    )
  return(anova_table)
}

#

mean_y = mean(training_data[,"Y"])

r_h0 = mean_y - training_data[,"Y"]

r_h1 = linear_model$residuals

r_r = predict(linear_model, training_data) - mean_y

anova_table = data.frame(row.names = c("Regression", "Error", "Total"))

anova_table = cbind(anova_table, "D.F."=c(
  1,
  nrow(training_data)-2,
  nrow(training_data)-1)
)

anova_table = cbind(anova_table, "S.S."=c(
  sum(r_r^2),
  sum(r_h1^2),
  sum(r_h0^2))
)

anova_table = cbind(
  anova_table,
  "M.S.S."= c(anova_table[,"S.S."]/anova_table[,"D.F."])
)

print(anova_table)
```

```
##            D.F.       S.S.    M.S.S.
## Regression    1   36078511 36078511
## Error        56 327710439   5851972
## Total        57 363788950   6382262
```

```
f_value = anova_table[["Regression", "M.S.S."]]/anova_table[["Error", "M.S.S."]]

p_value = 1-pf(f_value, 1, 56)
```

The F value is 6.1651884. The imputed P-value is 0.0160508.

This is confirmed by the built-in ANOVA function:

```
summary(aov(linear_model))
```

```
##             Df    Sum Sq  Mean Sq F value Pr(>F)
## X            1  36078511 36078511   6.165 0.0161 *
## Residuals   56 327710439  5851972
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

d-) *Carry out the test in part a by means of the T test.*

Is the P-value for the F test the same as that for the t test?

T-Test

The Student's t-test and the one-sided F-test (with 1 numerator degree of freedom) are equivalent when comparing two groups or testing a single regression coefficient.

$$F(1, df_d) = t^2(df_d)$$

The two-tailed T-Test and the one-tailed F-Test are related thusly:

$$F_{0.95}(1, n-2) \cong t_{.975}(n-2)^2$$

Show the numerical equivalence of the two test statistics and decision rules.

```
t_squared = summary(linear_model)$coefficients["X","t value"]^2
f = anova(linear_model)["X","F value"]

print(t_squared-f)
```

```
## [1] -8.881784e-16
```

The difference between t^2 and f is negligibly small.

```
critical_t_squared = qt(0.975, nrow(training_data)-2)^2
critical_f = qf(0.95, 1, nrow(training_data)-2)

print(critical_t_squared - critical_f)
```

```
## [1] -8.881784e-16
```

The difference_between $(t^*)^2$ and f^* is negligibly small.

e-) *By how much is the total variation in crime rate (Y) reduced when percentage of high school graduates (X) is introduced into the analysis?*

```r
print(anova_table)
```

```
##            D.F.      S.S.   M.S.S.
## Regression    1  36078511 36078511
## Error        56 327710439  5851972
## Total        57 363788950  6382262
```

The sum of squared residuals for the model without X is 3.6378895×10^8.

The sum of squared variation for the model with X is 3.2771044×10^8.

The total variation is reduced by 3.6078511×10^7

Is this a relatively large or small reduction?

```r
SST = anova_table["Total", "S.S."]
SSE = anova_table["Error", "S.S."]
coeff = (SSE - SST) / SST
percent_decrease = (1 - coeff) / 100
```

This constitutes a reduction of 0.0109917%. This is significant.

f-) *State the full and reduced models.*

The full model is

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

The reduced model is

$$Y = \beta_0 + \varepsilon$$

g-)

Obtain

1. *SSE(F),*
2. *SSE(R),*
3. *dfF.*
4. *dfR,*
5. *test statistic F* for the general linear test,*
6. *decision rule.*

These figures are equivalent to the figures in part B, and in the ANOVA table.

`print(anova_table)`

$SSE(F)$ is equivalent to SSE. $SSE(R)$ is equivalent to SST. dfF is equivalent to DFE. dfR is equivalent to DFT.

The test statistic $F*$ is 4.0129734.

The decision rule is to reject the null hypothesis if $SSR/MSE < F*$, else fail to reject.

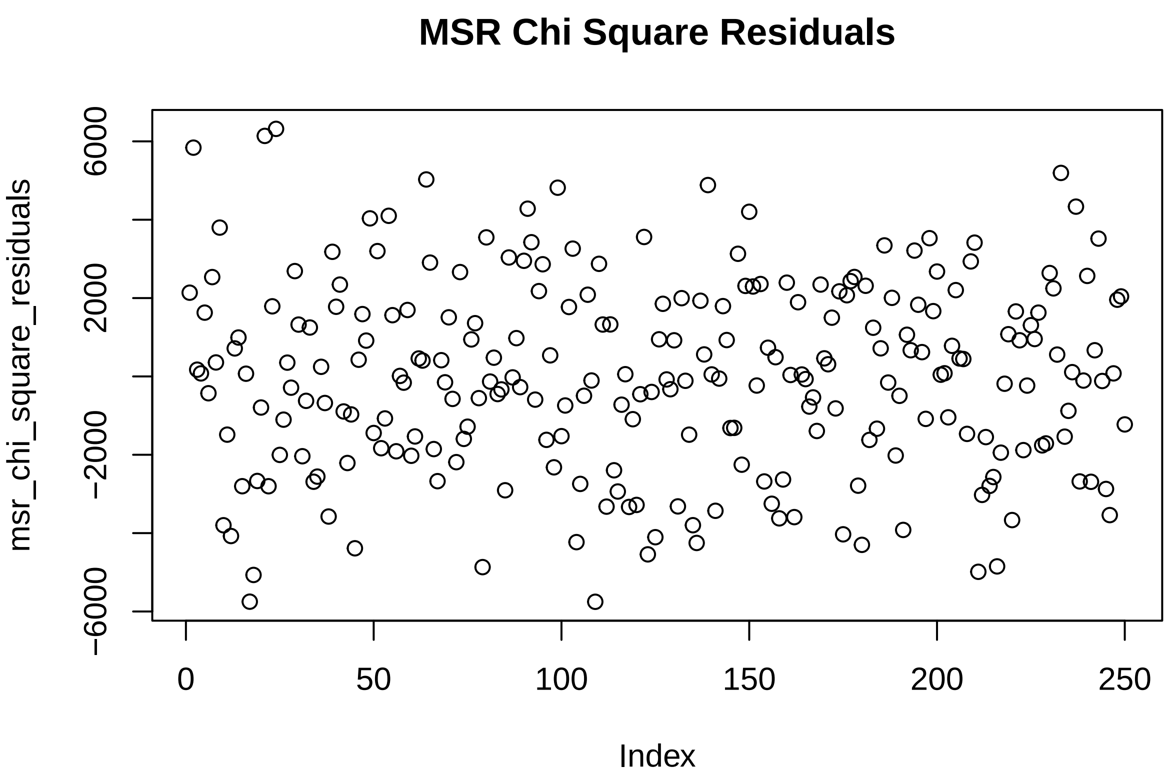
h-) Are the test statistic F* and the decision rule for the general linear test numerically equivalent to those in part b?

Yes.

i-) Evaluate model performance and stability on the train data set.

Let's run it over a bunch of samples.

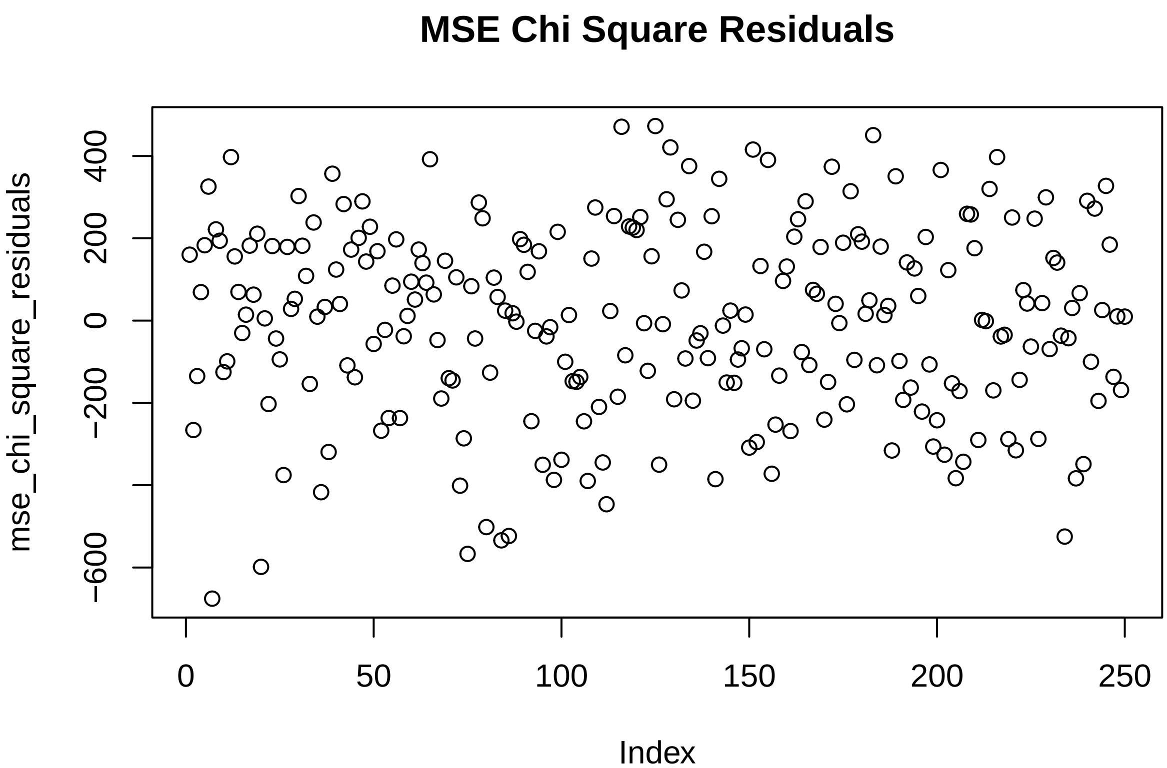
```
RUNS = 250
run_data = list()
for (i in 1:RUNS) {
  idx_training_data = sample(popN, size = alphaN, replace=FALSE)
  training_data = crime_rate_data[idx_training_data,]
  test_data = crime_rate_data[-idx_training_data,]

  # Run Models
  anova_table = anova_table_constructor(training_data)
  run_data[[i]] = data.frame(
      MSR=anova_table["Regression", "M.S.S."],
      MSE=anova_table["Error", "M.S.S."]
    )
}
run_data = do.call(rbind, run_data)

msr_chi_square_test = chisq.test(run_data$MSR)
msr_chi_square_residuals = msr_chi_square_test$residuals
plot(msr_chi_square_residuals, main="MSR Chi Square Residuals")
```




```
print(msr_chi_square_test)
```

```
##
##  Chi-squared test for given probabilities
##
## data:  run_data$MSR
## X-squared = 1457345641, df = 249, p-value < 2.2e-16
```

```r
mse_chi_square_test = chisq.test(run_data$MSE)
mse_chi_square_residuals = mse_chi_square_test$residuals
plot(mse_chi_square_residuals, main="MSE Chi Square Residuals")
```




```r
print(mse_chi_square_test)
```

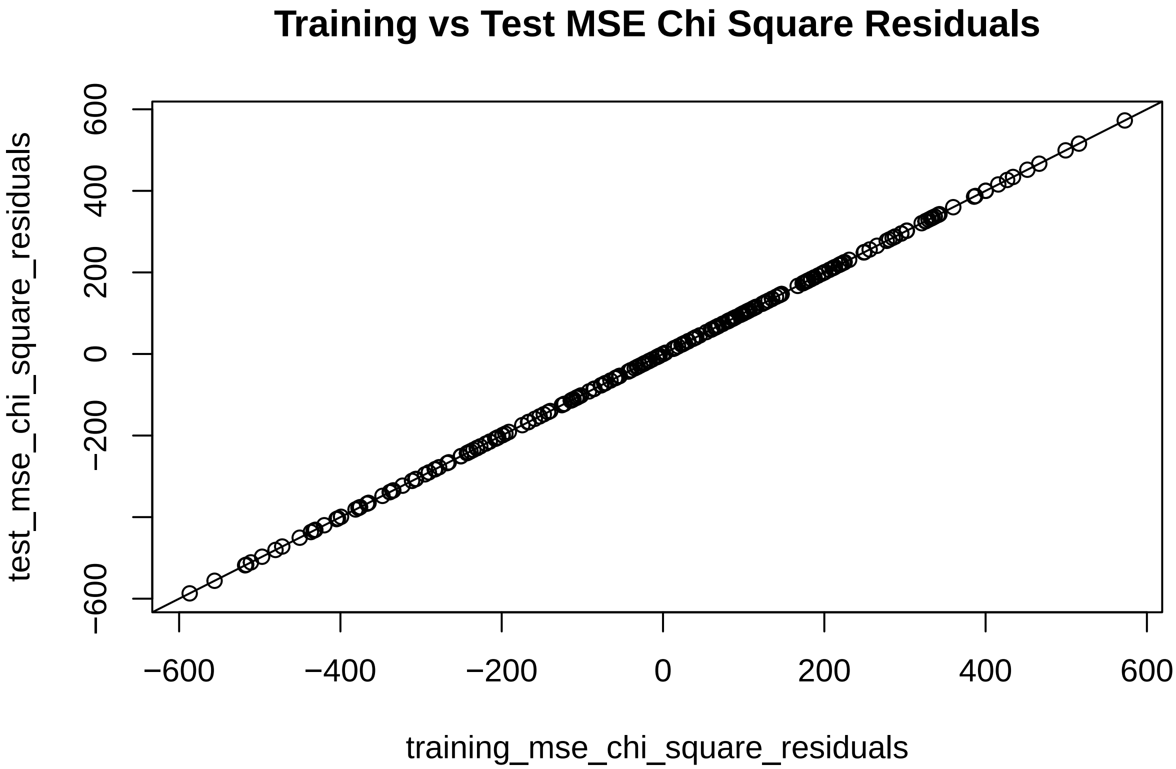
```
##
##  Chi-squared test for given probabilities
##
## data:  run_data$MSE
## X-squared = 13168860, df = 249, p-value < 2.2e-16
```

Use test data set below

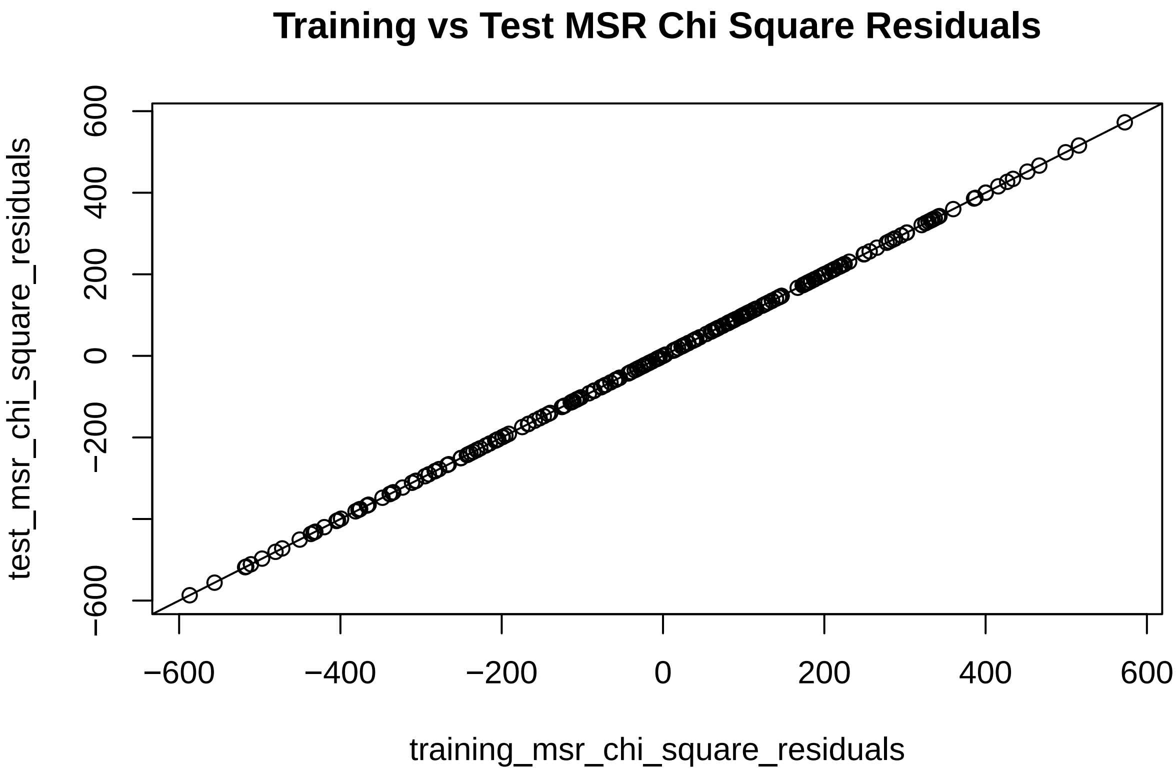
j-) Evaluate model performance and stability on the test set and compare it to the training set. Comment on your findings? (20 points)

```r
RUNS = 250
run_data = list()
for (i in 1:RUNS) {
  idx_training_data = sample(popN, size = alphaN, replace=FALSE)
  training_data = crime_rate_data[idx_training_data,]
  test_data = crime_rate_data[-idx_training_data,]

  # Run Models
  training_anova_table = anova_table_constructor(training_data)
  test_anova_table = anova_table_constructor(test_data)
  run_data[[i]] = data.frame(
      trainingMSR=training_anova_table["Regression", "M.S.S."],
      trainingMSE=training_anova_table["Error", "M.S.S."],
      testMSR=test_anova_table["Regression", "M.S.S."],
```

```
      testMSE=test_anova_table["Error", "M.S.S."]
    )
}
run_data = do.call(rbind, run_data)

training_mse_chi_square_test = chisq.test(run_data$trainingMSE)
training_mse_chi_square_residuals = training_mse_chi_square_test$residuals
test_mse_chi_square_test = chisq.test(run_data$trainingMSE)
test_mse_chi_square_residuals = test_mse_chi_square_test$residuals
plot(training_mse_chi_square_residuals, test_mse_chi_square_residuals, main="Training vs Test MSE Chi Sc
abline(0,1)
```




```
training_msr_chi_square_test = chisq.test(run_data$trainingMSE)
training_msr_chi_square_residuals = training_msr_chi_square_test$residuals
test_msr_chi_square_test = chisq.test(run_data$trainingMSE)
test_msr_chi_square_residuals = test_msr_chi_square_test$residuals
plot(training_msr_chi_square_residuals, test_msr_chi_square_residuals, main="Training vs Test MSR Chi Sc
abline(0,1)
```


Training vs Test MSR Chi Square Residuals
test_msr_chi_square_residuals
training_msr_chi_square_residuals
−600
−400
−200
0
200
400
600